

Polymer pretreatments for enhanced adhesion of metals deposited by the electroless process

M. Charbonnier, M. Romand*

Laboratoire de Sciences et Ingénierie des Surfaces, Université Claude Bernard, Lyon 1, 43 Boulevard du 11 Novembre 1918, 69622 Villeurbanne Cedex, France

Accepted 13 March 2003

Abstract

Metallization techniques based on electroless plating are widely used to coat polymer materials in a large variety of technological applications. Traditionally, dilute tin chloride (SnCl_2) and palladium chloride (PdCl_2) solutions in HCl are used to render the surface of such non-conductive substrates catalytically active towards metal deposition in the electroless plating solution. In the present work, it is shown how X-ray photoelectron spectroscopy has allowed to monitor the chemical and compositional surface modifications of polymer substrates (polypropylene, polycarbonate) subjected to plasma and UV or VUV irradiation (use of ArF^* excimer laser and Xe_2^* incoherent excimer lamp, respectively) in oxygenated (O_2 , air) or nitrogenated (N_2 , NH_3) atmospheres, as well as to understand the mechanisms of the catalyst (palladium species) chemisorption on the so-grafted surfaces through the use of a simple dilute palladium chloride solution in HCl. In addition, this work has allowed to bring into light the precise role that the reducer plays (sodium hypophosphite) present in the electroless nickel bath. In short, this research has been successful in allowing the development of new approaches for the electroless metallization of polymer surfaces.

© 2003 Elsevier Ltd. All rights reserved.

Keywords: B. Plastics; B. Surface treatment by excited gases; C. X-ray photoelectron spectroscopy; Electroless metallization

1. Introduction

Metallization of polymers and polymer-based materials is used today in a large variety of technological applications ranging from the fabrication of printed circuits in microelectronics to decorative coating in general manufacturing [1–7]. For such applications, polymer substrates are commonly metallized by the electroless deposition process, which typically involves surface treatments to improve adhesion, surface seeding with an electroless catalyst (generally a palladium-containing compound), then immersion in an electroless bath (generally a Ni or Cu solution). As largely recognized, conventional metallization techniques, which are based on electroless plating, are hampered by a number of difficulties because of the structural incompatibility between the substrate and metallizing material, in terms of chemical bonding and properties [8]. Hence, it is not surprising that the interfacial bond

strength is often a factor which limits the performances of the relevant systems under operating conditions. These limitations as well as environmental restrictions have created a continually growing demand for the development of new efficient and dependable processes.

In the present paper, special attention is focused, first of all, on the development of surface modification methods providing potentially advantageous alternatives with a view to attach the catalyst on the polymer surface through a covalent bonding mechanism. These methods are based on the use of a radio-frequency glow-discharge plasma, Nd:YAG laser (fifth harmonic at 213 nm) and incoherent excimer lamp (Xenon source delivering VUV photons at 172 nm) operating in specific atmospheres (oxygenated or nitrogenated environments) and on the chemisorption of the catalyst on the so-functionalized surfaces. Details of the chemical interactions of such catalysts with the grafted surfaces are investigated through the use of X-ray photoelectron spectroscopy (XPS) after each stage of the preactivation processes. Some experiments are also carried out in order to understand the post-activation mechanisms. Examples shown in this work are relative to

*Corresponding author. Tel.: +33-472-44-81-68; fax: +33-4-72-43-12-06.

E-mail address: maurice.romand@univ-lyon1.fr (M. Romand).

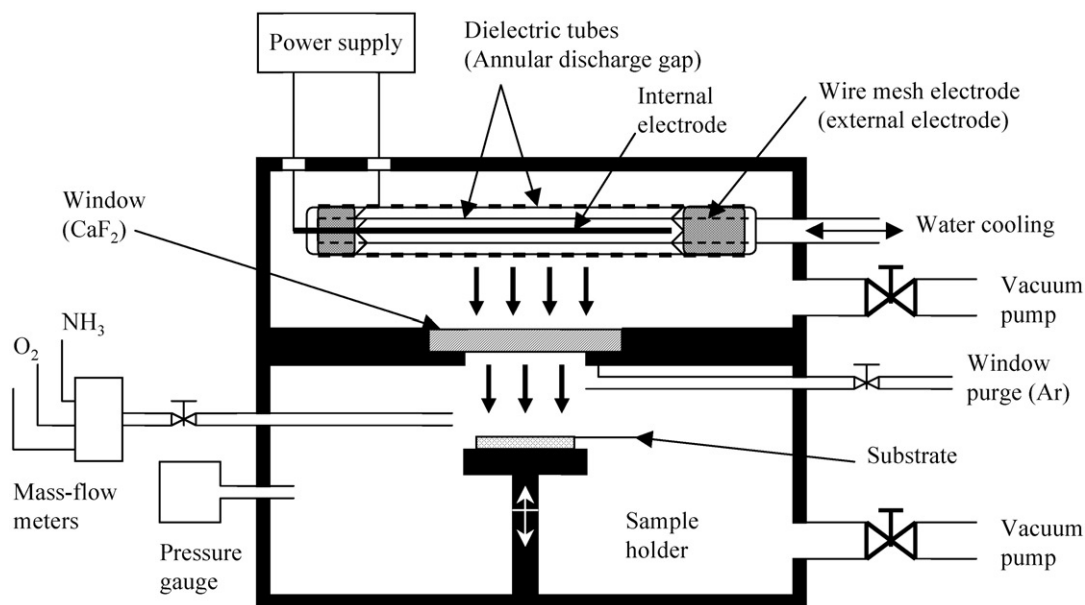
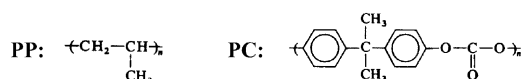


Fig. 1. Schematic diagram of the VUV irradiation system using an incoherent excimer lamp (Xe_2^*) as irradiation unit.

polypropylene (PP) and poly(bisphenol A carbonate) (PC) substrates and to electroless Ni plating.

2. Experimental

All surface pretreatments and subsequent Ni metallizations were performed on commercially available PP and PC foils obtained from Goodfellow, England. These samples were cleaned ultrasonically in ethanol before further processing.



Plasma processing was performed in a parallel plate RF reactor (diameter: 30 cm; distance between electrodes: 8 cm) operating in the reactive ion etching mode (RIE 80 reactor from Plasma Technology, England). The samples to be treated were placed on the RF powered electrode (17 cm in diameter) which was capacitively coupled, through an impedance matching network, to a 13.56 MHz RF power supply. The experimental conditions, unless specified otherwise, were as follows: gas (O_2 , N_2 , NH_3) flow = 100 sccm, working pressure = 100 mTorr, power density = 0.5 W cm^{-2} and variable treatment duration. The background pressure in the reactor was 1 mTorr.

Laser irradiation was performed with a Quantel Nd:YAG instrument delivering pulses in the picosecond range at a repetition rate of 10 Hz. In this work, the fifth harmonic ($\lambda = 213 \text{ nm}$, i.e. $h\nu = 5.82 \text{ eV}$, pulse duration 16 ps) was used. The energy density, in the range $0.10\text{--}0.80 \text{ mJ cm}^{-2}$, delivered by the laser beam on the sample surface was adjusted by varying the laser power.

The laser beam was focused on the polymer substrate (spot diameter = 7 mm) through a fused silica lens. The background pressure in the reaction chamber (a small laboratory-made reactor) was around 0.5 Torr. The distance between window and sample surface could be varied from 0.5 to 2 mm.

VUV irradiation was performed with a dielectric-barrier discharge (DBD) driven excimer lamp (Excivac Laboratory System, Heraeus Noblelight, Germany) operating in xenon and emitting an incoherent radiation peaking around $\lambda = 172 \text{ nm}$ ($h\nu = 7.20 \text{ eV}$). The DBD was operated at a frequency of about 275 kHz and at a voltage amplitude of 4 kV. In these experiments, the Xe_2^* excimer lamp was mounted in an experimental set-up which consists of two contiguous chambers (Fig. 1). The reactor itself, viz. the lower chamber, is separated from the upper chamber, in which the excimer lamp is placed, by a 5 mm thick calcium fluoride (CaF_2) window. In all cases, the upper chamber was pumped down ($\sim 10^{-2}$ Torr). When processing takes place in air at atmospheric pressure, the sample was located at a distance less than 1 cm from the CaF_2 window. On the other hand, when processing takes place in air or in another gas at reduced pressure, the sample was also placed at a distance of 1 cm from the window even though this distance is much less critical [9] than in the case of treatment at atmospheric pressure. Such experimental conditions have to be taken into consideration because oxygen molecules strongly absorb the VUV radiation and are dissociated. This leads to the formation of ozone (O_3) which also attenuates the radiation. The flux density generated by the lamp is estimated between 2.5 and $3.3 \times 10^{16} \text{ photons cm}^{-2} \text{ s}^{-1}$ (i.e. between 30 and 40 mW cm^{-2}) [10].

In this work, two methods were considered with a view to doing the functionalized polymer substrates “catalytically” active through the adsorption of palladium nuclei. The first method is based on the use of the conventional sensitization/activation process which consists of immersing the substrates for 2 min in a solution containing 0.1 g l^{-1} SnCl_2 and 0.1 ml l^{-1} HCl (37%), rinsing them in deionized water, immersing them for 2 min in a solution containing 0.1 g l^{-1} PdCl_2 and 3.5 ml l^{-1} HCl (37%), and rinsing. The second method involves a newly developed activation process which is carried out by immersing the substrates in a simple solution containing 0.1 g l^{-1} PdCl_2 and 3.5 ml l^{-1} HCl (37%) for 2 min, followed by rinsing.

Ni metallization was carried out at 85°C using the following home-made solution ($\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$: 36 g l^{-1} , $\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$: 10 g l^{-1} , lactic acid (85%): 29 ml l^{-1}) whose pH value was adjusted to 5 by NaOH pellets. Ni–P films were electrolessly deposited onto activated polymer surfaces.

Surface analysis was carried out using XPS with a Riber SIA 200 spectrometer equipped with a non-monochromatic twin Al/Mg excitation source and a MAC 2 semi-imaging analyser. In this work, characterization of the polymer substrates at different stages of their surface modification was performed using Al $\text{K}\alpha$ radiation. Spectra were recorded at a take-off angle of 65° with respect to the sample surface. Wide-scan and core-level spectra were recorded with an intrinsic analyser resolution of 2.2 and 1.0 eV, respectively. Given the natural width of the Al $\text{K}\alpha$ line, the resulting experimental resolution (i.e. the spectrometer resolution) is 2.3 and 1.3 eV, respectively. All XPS peaks were referenced to the C 1s signal, at a binding energy of 285 eV representative of the C–C and C–H bonds. Note that the use of a non-monochromatic source involves the presence of small satellite peaks due to Al or Mg $\text{K}_{\alpha 3,4}$ excitation radiation. Such satellites, which appear approximately at 10 and 12 eV on the low binding energy side of the main photoelectron peaks, obviously need not be considered in the present study. The chemical effects concerning the elements constituting the polymer were interpreted according to literature data [11]. Samples were exposed to laboratory atmosphere for less than 2 min between surface modification and introduction in the XPS chamber.

Adhesion measurements of the metal films to their polymer substrates were performed using a standard Scotch[®] tape test (cross-hatched tape test) with a 3M 2503710 Scotch[®]-tape.

3. Results and discussion

Fig. 2 represents a schematic diagram of the surface preparation processes which are conventionally used to

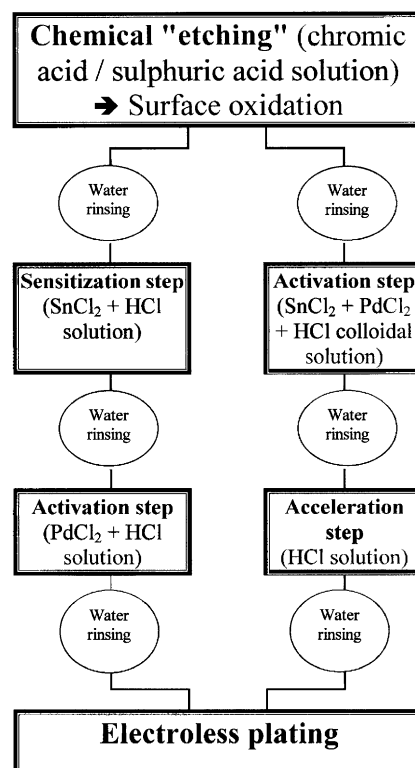


Fig. 2. Schematic diagram of the conventional electroless plating processes using a chemical etching pretreatment and the two-step sensitization/activation or one-step activation–acceleration procedures leading to the surface attachment of the catalyst (Sn/Pd “compound”).

make the polymer surfaces catalytically active [12]. After substrate cleaning with appropriate solvents to remove surface contaminants and chemical treatment (generally through chromic acid–sulphuric acid mixtures) to obtain a microroughened oxidized surface, the activation processes can be classified into two systems. The first system (historically speaking) involves the use of a sensitization step (immersion in a SnCl_2/HCl solution) followed by an activation step (immersion in a PdCl_2/HCl solution). The second system, the most widely employed system in practice today, involves the use of a direct activation step (immersion in a $\text{PdCl}_2/\text{SnCl}_2/\text{HCl}$ colloidal solution) followed by an acceleration operation (immersion in a HCl or NaOH solution). These two systems are often referred to as the two-step and one-step procedures, respectively. For more details concerning how these systems work, the reader is requested to consult [13,14] and references therein. In other respects, Fig. 3 represents the routes which were, at first, explored in our laboratory and which have led to the development of new and simplified electroless plating methods [13–19]. The latter make use of plasma or UV-laser-induced conditioning procedures instead of chemical etching processes used in the conventional methods.

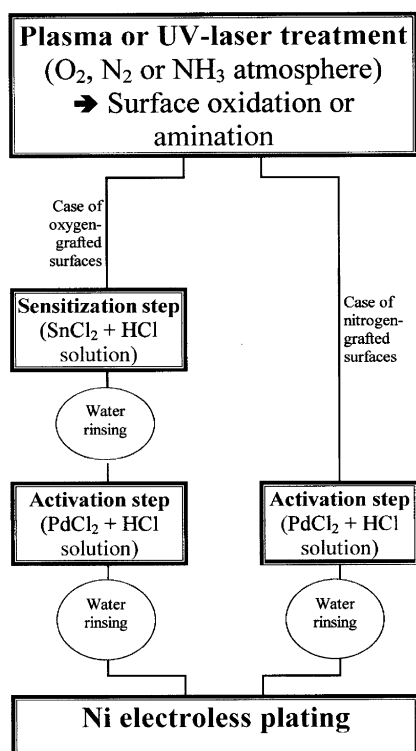


Fig. 3. Schematic diagram of the electroless plating processes using a plasma or UV-laser pretreatment in O_2 , N_2 or NH_3 atmosphere, and the two-step sensitization/activation (case of oxygen-grafted surfaces) or one-step direct activation (case of nitrogen-grafted surfaces) procedures leading to the surface attachment of the catalyst (Sn/Pd “compound” or Pd(+2) species, respectively).

3.1. Surface conditioning using plasma-induced procedures in oxygenated or nitrogenated atmospheres

Fig. 4 represents XPS wide-scan spectra of PP substrates after cleaning in ethanol (a) and after plasma treatment for 1 min in O_2 , N_2 and NH_3 atmospheres (spectra b, c and d, respectively). The O 1s peak observed on PP after ethanol cleaning is due to residual impurities in the substrate. An O_2 plasma leads to the chemisorption of oxygenated functionalities while N_2 and NH_3 plasmas cause the chemisorption of both oxygenated and nitrogenated functionalities. It is a common phenomenon that oxygen is incorporated on a polymer surface during and after non-oxygen plasma treatments. Free radicals that are created on a polymer surface can react with residual oxygen during a plasma treatment. Moreover, radicals that remain on a polymer surface after plasma treatment react with oxygen when the sample is brought to atmosphere. Fig. 5 (spectrum (b)) shows XPS wide-scan spectra of the so-grafted surfaces after immersion successively in a $SnCl_2$ acidic solution (sensitization step) and in a $PdCl_2$ acidic solution (activation step), i.e. after the two-step procedure. The O_2 plasma-treated sample strongly chemisorbs Sn and to a lower extent Pd. In the present case, a

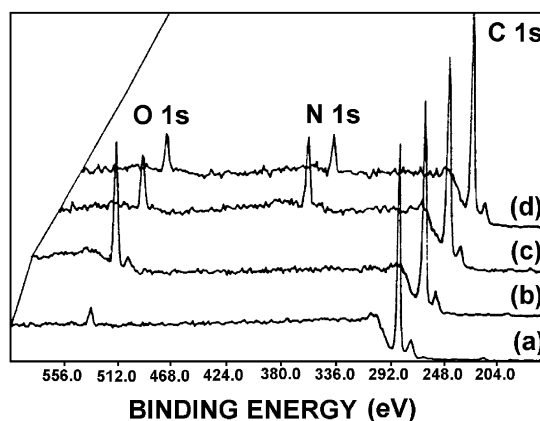


Fig. 4. XPS wide-scan spectra of PP substrates: (a) after cleaning (ethanol), (b), (c) and (d) same as (a) after plasma treatment in O_2 , N_2 and NH_3 atmospheres, respectively (working pressure = 100 mTorr, flow-rate = 100 sccm, treatment time = 1 min).

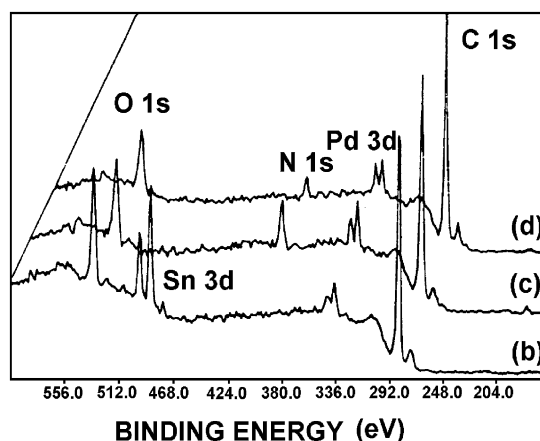


Fig. 5. XPS wide-scan spectra of O_2 (b), N_2 (c) and NH_3 (d) plasma-treated PP samples after the two-step sensitization/activation process (immersion successively in $SnCl_2$ and $PdCl_2$ solutions), and water rinsing.

Pd/Sn atomic ratio of 0.23 is measured by XPS. This result is in full agreement with those reported (range from 0.2 to 0.3) by different authors [20–22] and obtained by Mössbauer, X-ray fluorescence and Rutherford backscattering measurements. On the other hand, the N_2 and NH_3 plasma-treated samples (spectra c and d, respectively) do not attach any tin species but chemisorb a significant surface concentration of palladium species. In order to confirm that palladium has a marked propensity to be bonded to nitrogenated species rather than to oxygenated species, the grafted surfaces corresponding to samples b–d were immersed in a simple $PdCl_2$ acidic solution. As shown in Fig. 6, the surface only grafted with oxygenated species, practically does not adsorb any Pd (spectrum b) while those carrying nitrogenated species (spectra c and d) show a specific ability to chemisorb Pd (1.7 at. % in these experiments). The results described here above

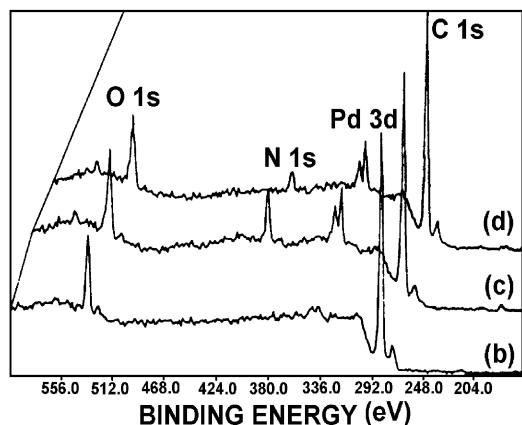


Fig. 6. XPS wide-scan spectra of O₂ (b), N₂ (c) and NH₃ (d) plasma-treated PP samples after the one-step activation process (immersion in a PdCl₂ solution), and water rinsing.

concerning PP substrates can be extended to many other polymers and polymer-base materials containing various heteroatoms (O, N, F, Cl, S, etc.). Clearly, these results can be summarized as follows:

- (i) Pd, the catalyst of the redox reaction, cannot be chemisorbed directly on polymer surfaces carrying only oxygenated species while Sn can. This can be explained by the strong chemical affinity of tin towards oxygen and by the non-affinity of palladium towards this same element.
- (ii) on the other hand, Pd can be directly chemisorbed on polymer surfaces carrying nitrogenated species. This can be explained by the strong chemical affinity of palladium towards nitrogen. In this case, a simplified route using only a palladium chloride acidic solution is enough to chemisorb the catalyst.

Modelling of the Pd and Sn species chemisorption on surfaces carrying nitrogenated and oxygenated functionalities can be found in a previous paper [16]. In the one-step tin-free process, Pd(+2) ion adsorption is only possible on surfaces carrying nitrogenated groups. The mechanism we proposed would consist in the formation of coordinate covalent bonds, the N atoms giving their lone-pair electrons to Pd ions which, in this way, complete their 4d electronic level and get a higher stability. In the two-step process using successively SnCl₂ and PdCl₂ solutions, Sn(+2) ions adsorb on oxygenated groups through the formation of Sn–O covalent bonds, i.e. through the sharing of their valence electrons. Then, Pd(+2) ions attach on tin species via a complex and not yet elucidated way. In both cases (one-step and two-step processes), it was also suggested that Cl[−] ions would participate in the formation of the surface “clusters” which contain the Sn and/or Pd species.

In addition, note that the high surface concentration of palladium species grafted, in particular, on NH₃

plasma-treated substrates provides a high number of nucleation sites for nickel deposition which suggests an adhesion improvement of the nickel deposits. Indeed, 2 μm thick Ni films deposited with the home-made Ni bath by the simplified one-step process pass the Scotch[®]-tape test, while similar Ni films deposited by the conventional two-step process do not pass it. Such a result was also obtained for other substrates and, in particular, for polytetrafluoroethylene (PTFE) by using a fragmentation test [13].

3.2. Surface conditioning using UV-laser-induced procedures in nitrogenated atmospheres

Considering the key role that UV ($\lambda > 200$ nm) and VUV ($\lambda < 200$ nm) radiations are able to play in the plasma processing of polymer surfaces and in the grafting of specific functionalities [23, 24], the conditioning of polymer substrates has also been carried out using a Nd:YAG laser. Such a laser can be employed for UV applications thanks to its capability of generating relatively high photon densities and thanks to the efficiency of optical crystals to produce frequency multiplication of the fundamental 1064 nm wavelength. In the present experiments, the fifth harmonic at 213 nm was used to treat the surface of polycarbonate substrates. Ammonia was used as nitrogenated atmosphere. Indeed, as far as photon energy is concerned, the radiation at 213 nm is energetic enough (5.82 eV) to break the N–H bonds (4.03 eV) of NH₃ molecules, but is not enough to break the N≡N bonds (9.76 eV) of N₂ molecules. In a similar way, this radiation is energetic enough to break many bonds at the polymer surface, and hence to form reactive free radicals.

Fig. 7 shows XPS wide-scan spectra of polycarbonate (PC) substrates (a) after cleaning in ethanol, and (b) after laser irradiation (total fluence = 4.68 J cm^{−2} corresponding to a fluence of 0.52 mJ cm^{−2} per shot for

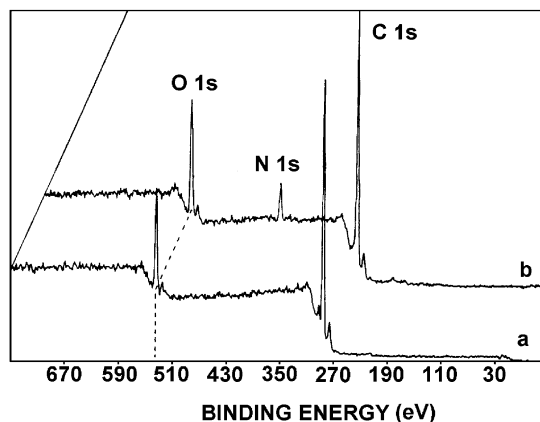


Fig. 7. XPS wide-scan spectra of polycarbonate substrates: (a) ethanol-cleaned, (b) same as (a) after laser irradiation in NH₃ at 7.5 Torr (total dose of irradiation: 4.68 J cm^{−2}).

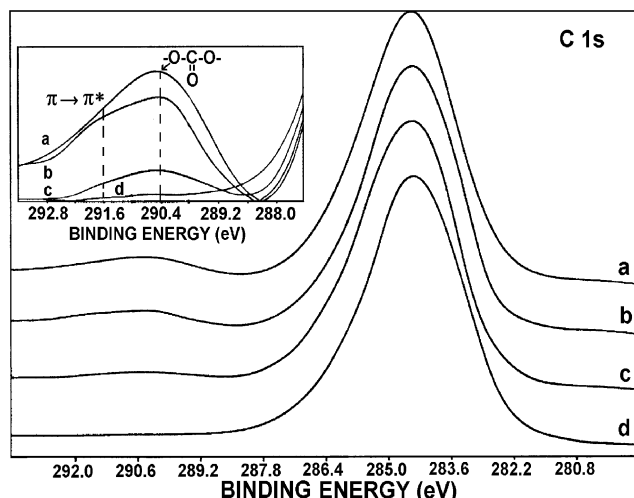


Fig. 8. C 1s XPS spectra of polycarbonate substrates: (a) ethanol-cleaned, (b), (c) and (d) same as (a) after laser irradiation in NH_3 at 7.5 Torr with a total fluence of 1.56, 4.68 and 6.24 J cm^{-2} , respectively. The inset represents the corresponding changes in the C 1s high-energy “satellite”.

15 min) in ammonia (NH_3) at reduced pressure (7.5 Torr). In this example, the carbon surface concentration decreases from 87.9 to 79.2 at% while that of oxygen increases from 12.1 to 15.5 at%. In addition, the surface concentration in nitrogen grafted during the laser irradiation reaches 5.3 at%. Under the experimental conditions used in this work, the PC substrate appears to be less strongly modified by the laser treatment than by a plasma treatment [16,17]. A first interesting result concerning the surface modification of PC substrates during the UV irradiation in the NH_3 atmosphere can be extracted from the C 1s XPS peak. Fig. 8 represents C 1s spectra of PC substrates (a) after cleaning in ethanol, (b), (c) and (d) after a subsequent irradiation with a total fluence of 1.56, 4.68 and 6.24 J cm^{-2} , respectively. These spectra were normalized in order to observe, in particular, the changes in intensity and shape of the high-energy peak (see insert in Fig. 8), due for one part to the C 1s peak of the carbonate group (290.4 eV) and for another part to the bonding–antibonding ($\pi \rightarrow \pi^*$) transition (291.6 eV) revealing the presence of the aromatic rings. The intensity of these components decreases progressively as the total fluence of irradiation increases. This means that the laser exposure causes the opening of the surface aromatic rings and disappearance of the carbonate functionalities, probably through C–O bond breaking.

Complementary information on the surface modifications generated during the laser irradiation can be inferred from the O 1s peak. Fig. 9 represents O 1s spectra of the PC substrates (a) after cleaning in ethanol, (b), (c) and (d) after a subsequent irradiation in a NH_3 atmosphere (7.5 Torr), the total fluence of irradiation

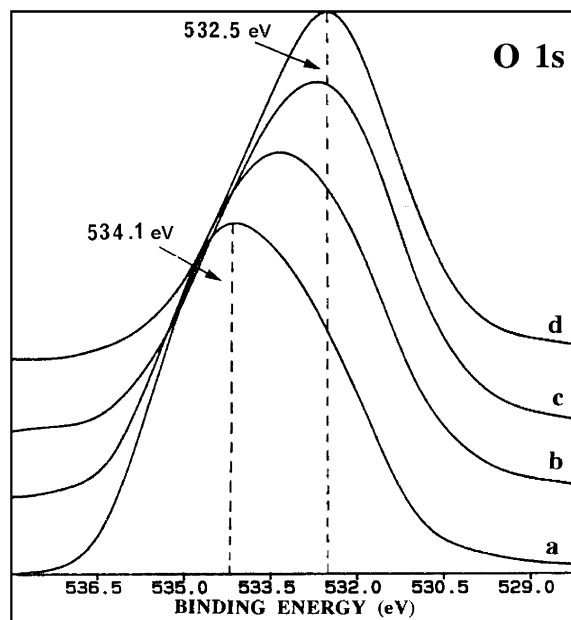


Fig. 9. O 1s XPS spectra of polycarbonate substrates (a) ethanol-cleaned, (b), (c) and (d) same as (a) after laser irradiation in NH_3 at 7.5 Torr with a total fluence of 1.56, 3.12 and 4.68 J cm^{-2} , respectively.

being 1.56, 3.12 and 4.68 J cm^{-2} , respectively. The O 1s peak is progressively shifted towards the low binding energy side as the total fluence increases. The O 1s spectrum characteristic of the unmodified sample includes the contribution of the two different chemical environments of oxygen atoms in carbonate groups: C–O–C bonds at 534.0 eV and O=C bonds at 532.3 eV [11]. The asymmetry of this spectrum is due to the fact that the number of C–O bonds is twice as important as that of C=O bonds. As the O 1s peak shifts towards a lower binding energy corresponding to oxygen atoms characteristic of O=C bonds, this means that the UV-laser irradiation causes the breaking of the C–O bonds. As a result, the so-formed active carbon sites become available to attach nitrogen atoms from the gas phase as it was the case after the breaking of the C–C and C–H bonds from the aromatic rings and methyl groups. The formation of amino groups is evidenced from the N 1s peak at 399.8 eV. It should be borne in mind that the C=O double-bond of the carbonate groups (bond strength $\sim 7.7 \text{ eV}$) cannot be broken by the UV photons delivered by the laser source used in this work. In addition, it is interesting to remark that excimer lasers providing UV or VUV photons can also be used to cause the “nitridation” (amination) of polymer surfaces. For example, an ArF* excimer laser ($\lambda = 193 \text{ nm}$, pulse duration = 20 ns) was employed for this purpose in the case of PTFE substrates [13]. In this case also, the energy of the photons delivered (6.4 eV) is higher than that of the N–H (4.03 eV) or C–F bonds (5.02 eV), so that nitrogenated functionalities can be grafted on the

PTFE surface. However, concurrent with the attachment of functional groups, some other phenomena including surface degradation and formation of weak boundary layers may occur during laser irradiation, depending on the experimental conditions. In the work reported here concerning PC substrates, the picosecond Nd:YAG laser was used because of its capability, in comparison with excimer lasers, of modifying the polymer surface in a more easily controlled manner, the energy density deposited per pulse, therefore per time unit being much lower.

Irrespective of these considerations, it must be concluded that grafting of nitrogenated species by appropriate laser irradiation through an ammonia atmosphere allows a further electroless metallization (Ni in the present study) via activation of the grafted surface in a simple PdCl_2 solution. For example, it was shown [18] that for different fluences and irradiation times, when the PC substrate is grafted with a nitrogen surface concentration between 5 and 6 at%, Pd is chemisorbed at a surface concentration of 1.0–1.1 at%. An additional interest of lasers is obviously their capability of treating limited areas selectively with a view to carry out localized metallizations. This way was used by Niino and Yabe [25, 26]. These authors were successful (i) in rendering the surface of fluoropolymers (PTFE and FEP) hydrophilic through the photolysis of hydrazine (N_2H_4) vapour during an ArF^* excimer laser irradiation (irradiation carried out through a mask), and (ii) in obtaining selective-area electroless plating (Ni or Cu) by means of sequences involving, among other steps, the predipping of the hydrophilic regions into an aqueous H_2SO_4 solution and the activation of the so-modified surface (photo-ablated and oxidized surface) via a conventional two-step process ($\text{Sn}(+2)/\text{Pd}(+2)$ route).

In this series of experiments, note that no adhesion test was made given the reduced size of the coated surfaces.

3.3. Surface conditioning using VUV-incoherent excimer lamp-induced procedures in nitrogenated atmospheres

In a first series of experiments, polycarbonate (PC) substrates were again considered. These substrates were surface-treated in oxygenated (air) and nitrogenated (ammonia) atmospheres (working pressure = 0.75 Torr) in the reactor described in Fig. 1. The samples were irradiated with the VUV photons emitted by the Xe_2^* incoherent excimer lamp after passing through the CaF_2 window. The working distance between this window and the sample holder was 1 cm. Fig. 10 represents the XPS wide-scan spectra of PC samples, respectively treated (a) in NH_3 , (c) in air for 3 min and of these same samples after a subsequent immersion in a simple PdCl_2 solution (one-step process) (spectrum (b)), and in $\text{SnCl}_2/\text{PdCl}_2$ solutions (two-step process) (spectrum (d)), respectively.

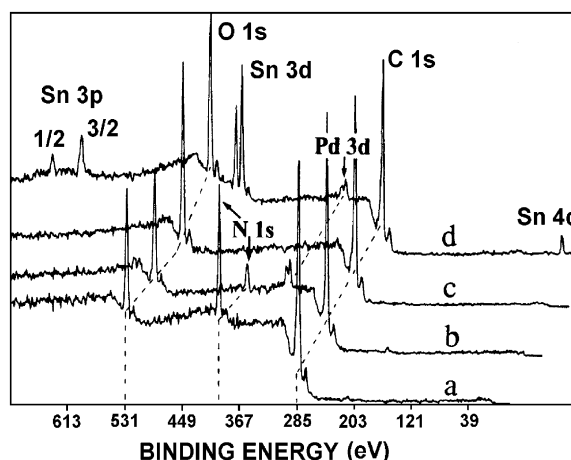


Fig. 10. XPS wide-scan spectra of ethanol-cleaned polycarbonate substrates after VUV treatment (Xe_2^* excimer lamp) in NH_3 (a) and air (c) at 0.75 Torr for 1 min, same as (a) after immersion in a simple PdCl_2 acidic solution (b), same as (b) after immersion successively in the $\text{SnCl}_2/\text{PdCl}_2$ acidic solutions (d).

As already pointed out by using plasma and/or UV-laser pretreatments, tin is strongly chemisorbed (spectrum (d)) on oxygen-grafted surfaces while palladium (spectrum (b)), on nitrogen-grafted surfaces. In other words, incoherent excimer sources are also quite able to render the polymer surface catalytically active for a subsequent electroless metallization. In the present series of experiments, nitrogen and palladium are grafted at surface concentrations of 6.1 and 0.9 at%, respectively, when the PC substrates are pretreated in an ammonia environment. On the other hand, when these same substrates are pretreated in air at reduced pressure, tin is grafted (onto oxygenated species) at a surface concentration of about 3.1 at% while palladium is at a surface concentration of only 0.8 at%. As in the case of NH_3 plasma treatments, the processing using grafting of nitrogenated species through the use of the VUV excimer lamp is very performant in view to attaching a high surface density of catalyst, which let also envisage the formation of a stronger electroless metal/polymer substrate interface. This is confirmed by using the Scotch[®] tape test.

3.4. Chemical state of the active catalyst

Whatever are the procedures used to attach palladium species onto polymer substrates, the determination of the palladium state after the conventional two-step sensitization/activation or one-step activation–acceleration sequences has been the subject of numerous studies. Controversial results regarding both various substrates and activation modes [20–22, 27–31] have been described in the literature. They show on the one hand that palladium is adsorbed or complexed as $\text{Pd}(+2)$, and on the other hand that it is present as $\text{Pd}(0)$. In the present work, evidence is shown that the sensitized and

activated surfaces, as well as the directly activated surfaces become active for Ni deposition only after a certain immersion time in the electroless bath (about 2 s in the first case, 15 s in the second case). In other words, it is brought into light that palladium is deposited as Pd(+2), and that the plating bath itself is capable of reducing these species into Pd(0) species, which serve to nucleate the metal deposition from the electroless plating bath. For example, Fig. 11 represents Pd 3d XPS spectra of polycarbonate substrates. Spectrum (b) is relative to a PC sample plasma-treated in NH_3 and then immersed in a PdCl_2 solution while spectra (c), (d) and (e) are characteristic of the same sample as (b), respectively after immersion in the plating bath for 12 s (spectrum (c)), after immersion in an acidic aqueous solution containing only the reducer (sodium hypophosphite) for 12 s (spectrum (d)) or 10 min (spectrum (e)). The comparison with the Pd 3d reference spectra of metallic Pd (spectrum (f)) and Pd(+2) species from a PdCl_2 compound (spectrum (a)) clearly shows that the

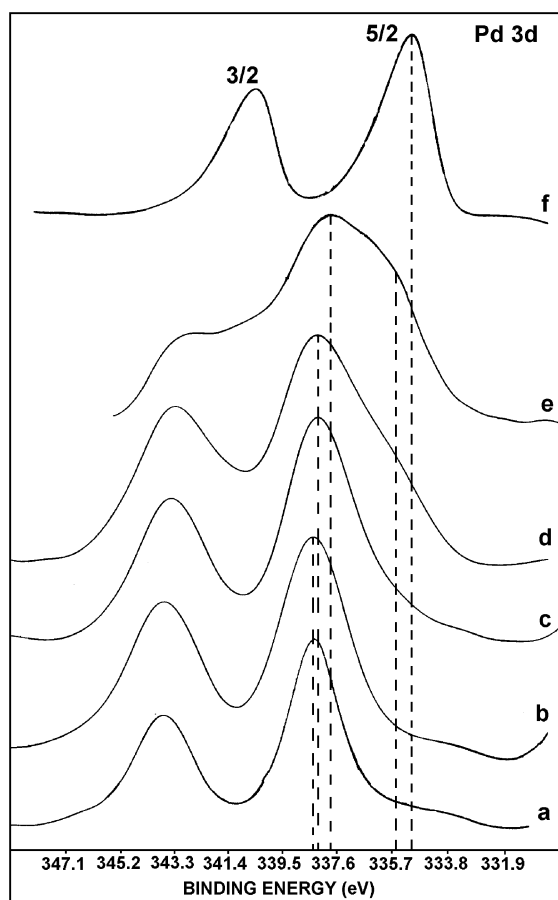


Fig. 11. Pd 3d 3/2, 5/2 XPS spectra of polycarbonate substrates plasma-treated in NH_3 (100 mTorr, 1 min) and then dipped into a PdCl_2 solution (b), same as (b) after immersion in the Ni electroless bath for 12 s (c), same as (b) after immersion in a NaH_2PO_2 , H_2O acidic solution for 12 s (d) and 10 min (e), respectively. The Pd 3d 3/2, 5/2 spectra of metallic palladium (f) and solid PdCl_2 (Pd(+2) species) (a) are given for comparison.

shoulder appearing towards the low binding energy side of each spin doublet peak is due to Pd(0) species which are formed in the plating bath during the incubation time thanks to the action of the reducer.

4. Conclusions

XPS is undoubtedly the most useful and widely used surface analysis technique available both in academic laboratories and research and development (R&D) industrial facilities. Throughout the past 30 years, many applications have been reported in the literature and the considerable potential of this tool, in particular, in the study of polymer surfaces has been brought into light. The present work describes a meaningful additional example showing how XPS analysis has provided some information which advances our understanding of fundamental mechanisms governing the formation of the metal–substrate interface in the electroless plating of polymer materials. In short, XPS measurements have aided in the identification and quantification of (i) the chemical functionalities grafted on various polymer surfaces subjected to plasma or UV/VUV irradiation (use of ArF^* excimer laser and Xe_2^* incoherent excimer lamp) in oxygenated and nitrogenated atmospheres, (ii) the specific chemical interactions occurring between the sensitization and/or activation baths (dilute SnCl_2 and/or PdCl_2 solutions) and the so-grafted species, namely:

- the strong chemical affinity of tin and the non-affinity of palladium towards oxygenated species,
- the strong chemical affinity of palladium and the non-affinity of tin towards nitrogenated species.

In addition, XPS measurements have allowed to clarify the role of the reducer (sodium hypophosphite) present in the nickel plating bath. Indeed, the “activated” surface becomes catalytically active for electroless deposition only after a certain immersion time during which a sufficient surface density of the chemisorbed Pd(+2) species are converted into Pd(0) nuclei. Finally, it is important to emphasize, from an experimental point of view, that these studies have led to the development of new approaches for the electroless plating of polymer surfaces.

References

- [1] Mittal KL, Susko JR. Metallized plastics 1: fundamental and applied aspects. New York: Plenum Press; 1989.
- [2] Sacher E, Pireaux JJ, Kowalczyk SP. Metallization of polymers. Washington, DC: American Chemical Society; 1990.
- [3] Mittal KL. Metallized plastics 2: fundamental and applied aspects. New York: Plenum Press; 1991.
- [4] Mittal KL. Metallized plastics 3: fundamental and applied aspects. New York: Plenum Press; 1992.

- [5] Mittal KL. Metallized plastics: fundamental and applied aspects. New York: Marcel Dekker; 1998.
- [6] Mittal KL. Metallized plastics 5 & 6: fundamental and applied aspects. Utrecht, The Netherlands: VSP; 1998.
- [7] Mittal KL. Metallized plastics: fundamental and applied aspects. Utrecht, The Netherlands: VSP; 2000.
- [8] Sun BK, Cho DL, O'Keefe TJ, Yasuda H. In: Mittal KL, Susko JR, editors. Metallized plastics 1: fundamental and applied aspects. New York: Plenum Press; 1989. p. 9–27.
- [9] Romand M, Charbonnier M. Functionalization of polymer surfaces with a new excimer UV/VUV source. Application to electroless metallization. In: Speth DR, editor. Proceedings of the 22nd Annual Meeting of the Adhesion Society, Blacksburg, VA, USA, The Adhesion Society Inc., 1999. p. 94–7.
- [10] Charbonnier M, Romand M, Esrom H, Kogelchatz U, Seeböck R. New approaches for electroless plating processes by activation of polymer surfaces using low pressure plasma and dielectric-barrier discharge devices. In: Mittal KL, editor. Metallized plastics 7: fundamentals and applied aspects. Utrecht, The Netherlands: VSP; 2001. p. 3–26.
- [11] Beamson G, Briggs D. High resolution XPS of organic polymers. The Scienta ESCA 300 database. Chichester: Wiley; 1992.
- [12] Mallory GO, Hajdu JB. Electroless plating: fundamentals and applications. Orlando, FL: American Electroplaters and Surface Finishers Society; 1990.
- [13] Romand M, Charbonnier M, Alami M, Baborowski J. Electroless metallization of polymers: simplification of the process by using plasma or UV-laser pretreatment. In: Mittal KL, editor. Metallized plastics 5 & 6: fundamental and applied aspects. The Netherlands: Utrecht VSP; 1998. p. 3–23.
- [14] Charbonnier M, Romand M, Stremdoerfer G, Fares-Karam A. Electroless metallization of polymers. *Recent Res Dev Macromol Res* 1999;4:27–43.
- [15] Alami M, Charbonnier M, Romand M. Interest of NH_3 and N_2 plasmas for polymer surface treatment before “electroless” metallization. *Plasmas Polym* 1996;1:109–22.
- [16] Charbonnier M, Alami M, Romand M. Plasma treatment process for palladium chemisorption onto polymers before electroless deposition. *J Electrochem Soc* 1996;143:472–780.
- [17] Alami M, Charbonnier M, Romand M. Plasma chemical modification of polycarbonate surfaces for electroless plating. *J Adhesion* 1996;57:77–90.
- [18] Charbonnier M, Alami M, Romand M, Girardeau-Montaut J, Afif M. Laser-assisted grafting onto polycarbonate: application to metallization by chemical means. *Appl Surf Sci* 1997;109/110: 206–11.
- [19] Charbonnier M, Alami M, Romand M. Electroless plating of polymers: XPS study of the initiation mechanisms. *J Appl Electrochem* 1998;28:449–53.
- [20] Cohen RL, West KW. Solution chemistry and colloid formation in the tin chloride sensitizing process. *J Electrochem Soc* 1972;119:433–8.
- [21] Cohen RL, D'Amico JF, West KW. Mössbauer study of tin II sensitizer deposits on Kapton. *J Electrochem Soc* 1971;118: 2042–6.
- [22] Meek RL. A Rutherford scattering study of catalyst systems for electroless Cu plating. II. SnCl_2 sensitization and PdCl_2 activation. *J Electrochem Soc* 1975;122:1478–81.
- [23] Liston EM. Plasma and surfaces—a practical approach to good composites. In: Akovali G, editor. The interfacial interactions in polymeric composites. Dordrecht The Netherlands: Kluwer Academic Publishers; 1993. p. 223–68.
- [24] Liston EM. Plasma modification of polymers surfaces. In: Pireaux JJ, Bertrand P, Bredas JL, editors. Polymer solid interfaces. Bristol, Great Britain: IOP Publishing Ltd; 1992. p. 429–42.
- [25] Niino H, Yabe A. Surface modification and metallization of fluorocarbon polymers by excimer laser processing. *Appl Phys Lett* 1993;63:3527–9.
- [26] Niino H, Murao T, Matsumura S, Yabe A. Hydrophilic surface modification and metallization of poly(tetrafluoroethylene) film by excimer laser processing. *Mol Cryst Liq Cryst* 1995; 267:365–74.
- [27] Sard R. The nucleation, growth and structure of electroless copper deposits. *J Electrochem Soc* 1970;117:864–70.
- [28] Chow SL, Hedgecock NE, Schlesinger M, Rezek J. The role of UV light in the inhibition of electroless deposition. *J Electrochem Soc* 1972;119:1013–8.
- [29] de Minjer CH, van der Boom PFJ. The nucleation with SnCl_2 – PdCl_2 solutions of glass before electroless plating. *J Electrochem Soc* 1973;120:1644–50.
- [30] Kiflawi I, Schlesinger M. Tin (IV)-based photoselective metal deposition of cobalt and nickel from alkaline baths. *J Electrochem Soc* 1983;130:872–4.
- [31] Baylis BKW, Busuttill A, Hedgecock NE, Schlesinger M. A study of the products of ultraviolet irradiation of palladium-containing catalysts for electroless metal deposition. *J Electrochem Soc* 1976;123:348–51.